

DTSA 5512

Introduction to Computer Vision

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (86.2 / 100)	74.1 / 100
SPECIALIZATION Computer Vision		

74

CURRENT AVG SCORE
across 6 job roles

100

POTENTIAL SCORE
with recommended additions

+34

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course guides students through the essential algorithms and methods to help computers 'see' and interpret visual data. Students learn the core concepts and techniques that have been traditionally used to analyze images. Then, students learn modern deep learning methods, such as neural networks and specific models designed for image recognition, can be used to perform more complex tasks like object detection and image segmentation. Additionally, students will learn the creation and impact of AI-generated images and videos, exploring the ethical considerations of such technology.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	86.2	100.0	+13.8	LOW
Machine Learning Engineer	84.9	100.0	+15.1	MEDIUM
ML Engineer	72.7	100.0	+27.3	MEDIUM
Applied Scientist	68.8	100.0	+31.2	HIGH
Data Scientist	66.3	100.0	+33.7	HIGH
MLOps Engineer	65.9	100.0	+34.1	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Business Communication 2 of 6 roles

insights

business

Deep Learning 2 of 6 roles

cnn

training

LLMs & Generative AI 2 of 6 roles

storage

Experiment Tracking 1 of 6 roles

tracking

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.

3 **MEDIUM** **Consider Deep Learning content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: cnn, training.

4 **MEDIUM** **Consider LLMs & Generative AI content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: storage.

5 **LOW** **Optional: Experiment Tracking** — would broaden appeal to one additional role. Candidate terms: tracking.

GAP KEYWORDS BY ROLE

AI Engineer

86.2 → 100.0 (+13.8 pts)

build

software

technical

systems

based

problems

storage

machine

Machine Learning Engineer

84.9 → 100.0 (+15.1 pts)

sets

business

design

science

applications

machine

performance

technical

ML Engineer

72.7 → 100.0 (+27.3 pts)

cnn

machine

systems

world

build

problems

product

new

Applied Scientist

68.8 → 100.0 (+31.2 pts)

- research
- storage
- literature
- diffusion
- problems
- technical
- systems
- industry

Data Scientist

66.3 → 100.0 (+33.7 pts)

- rules
- sets
- product
- mitigation
- insights
- build
- problems
- approaches

MLOps Engineer

65.9 → 100.0 (+34.1 pts)

- text
- systems
- machine
- tracking
- training

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.